Ibrahim Joudah

Data Scientist, Statistician & Applied AI Specialist

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Professional Profile

- Data scientist and statistician who has passed the PhD examination in Statistics, with industry experience translating complex data into reliable models, analytical tools and actionable business insights. Combines advanced statistical methodology with applied AI, including LLMs for predictive modelling and generative AI (GenAI) for synthetic-data generation. Designs and guides AI-assisted workflows to accelerate development, applying independent technical judgement to identify errors, refine solutions and rigorously validate results. Advanced R user who also works with Python, SQL, Power BI, Tableau and Excel. Communicates technical findings clearly to decision-makers and non-technical stakeholders.

Technical Skills

Programming & tools	R (advanced), Python, SQL, Excel, Tableau, Power BI, MATLAB, Alteryx, R Markdown and LaTeX.
Applied AI & GenAI	Agentic AI, large language models (LLMs) and GenAI, LLM evaluation, prompt design and structured outputs, natural language processing (NLP), text embeddings, synthetic-data generation, AI-assisted development and human-in-the-loop evaluation.
Statistics & ML	Statistical modelling, Bayesian modelling, regression, classification, predictive modelling, machine learning (XGBoost), deep learning, feature selection, anomaly detection, forecasting, simulation, experimentation, model evaluation and validation.
Software & workflows	Data pipelines, automation, production deployment (Linux, cloud), CI/CD and Git, monitoring, R package development, reproducible analysis, data preparation and validation, DataSHIELD and privacy-preserving federated analysis.
Communication	Technical reporting, stakeholder presentations, research publication, university teaching and translating complex analysis for non-technical audiences.

Professional Experience

Mar 2026 – present
Consultant, Pacific Analytics Consulting Pty Ltd Sydney, Australia

- Develop DataSHIELD-based R functions and packages for privacy-preserving statistical analysis across distributed datasets (federated analysis).
- Adapt statistical procedures to client-server constraints, including research and development of federated survival analysis methods, with testing and technical documentation.

Aug 2025 – Jun 2026
Quantitative Data Lead, Altamir AI & Data Governance Sydney, Australia

- Led quantitative analysis, study design, data-quality assessment, reconciliation and reporting for external AI and data-governance projects.
- Translated technical findings into practical recommendations for managers and senior stakeholders to support AI governance decisions.
- Supervised a multidisciplinary team of six UTS students through a three-month AI-governance project for the media sector, supporting survey design, recruitment, analysis, reporting and presentation.

Professional Experience (continued)

Feb 2023 – present
Teaching Associate, Macquarie University Sydney, Australia
- Teach undergraduate and postgraduate statistics and machine learning through practical classes, workshops, consultations and assessment using R and Excel.
- Develop clear explanations and visual resources for difficult concepts; student evaluations consistently recognise the clarity and depth of my teaching.

Dec 2022 – Jan 2023
Research Assistant, Environment Protection Authority Victoria Australia
- Contributed to experimental design, implementation, data analysis and reporting for a project estimating the number of tyres in stockpiles.

Jan 2019 – Aug 2019
Data Analyst Intern, Quant Data & Analytics Riyadh, Saudi Arabia
- Contributed to Fruits360, a SaaS platform for restaurant sales, operations and inventory analytics, through data preparation and joining, R integration, forecasting and Tableau dashboards.
- Developed automated anomaly-detection and robust data-handling approaches for outliers and data-entry errors.

Education

2022 – 2026
Doctor of Philosophy in Statistics, Macquarie University Sydney, Australia
Passed examination with the outcome "Award with corrections" (October 2026).
Thesis: *Feature Screening for Regression Models in Ultra-High Dimensional Settings*.

2021 – 2022
Master of Research in Statistics, Macquarie University Sydney, Australia
Thesis: *The ROPE Framework for Model Selection in Linear Regression*.

2016 – 2020
Bachelor of Science in Actuarial Science
King Fahd University of Petroleum & Minerals Dhahran, Saudi Arabia

Selected Projects, Research & Software

Summary: **6** Publications – **102** Citations – **6** Conference Presentations

2026
Agentic AI job-market monitoring system.
- Built and deployed an end-to-end agentic AI system in Python and SQL on a cloud server, running 24/7: it collects job ads from multiple sources, uses GenAI / LLMs to screen and evaluate each one, and sends notifications to phone.
- Designed the LLM evaluation: compared the AI's ratings with human judgements, so its decisions can be trusted and improved over time.
- Added a machine-learning model (XGBoost) that reduces paid AI calls; it was tested safely in shadow mode before switching on.
- Used statistical modelling (Bayesian) to keep monitoring fast while keeping server load low.
- Production-grade: about 1,300 automated tests, CI/CD with Git, monitoring and alerts, cost tracking, and encrypted backups.

StepAlytics: activity analytics and data science.
- Built two interactive Tableau dashboards supported by Python and SQL (DuckDB) workflows to explore activity patterns, weather relationships and program outcomes using activity data (e.g., step count) and real historical weather.
- Developed data-quality checks, statistical models, program evaluation and time-series forecasts; produced reports that explain findings, uncertainty and limitations.

Interactive Tableau dashboards GitHub: analysis notes and report

Selected Projects, Research & Software (continued)

- **Applied AI and predictive modelling for consumer surveys.**
  - Built a predictive modelling pipeline in R and Python combining LLM-generated estimates, text embeddings and XGBoost to predict survey qualification rates for categories with no labelled history.
  - Reduced prediction error (RMSE) by 21% relative to the calibrated LLM baseline, evaluated through cross-validation.
  - Produced uncertainty and recruitment estimates, automated reports, and tools for what-if analysis across markets and survey designs.

2024 – 2025 ● **Air-HOLP and AirScreen.** Developed an original feature-screening method for high-dimensional data with strongly related predictors, evaluated it through simulation and real-data studies, and translated it into the **AirScreen** R package published on CRAN. Published in *Statistics and Computing*.

2025 – 2026 ● **ROSE and ROSEScreen.** Developed a robust ensemble feature-screening framework for nonlinear relationships, outliers, errors and heavy-tailed noise, with reusable R software implementing the framework.

Publications ● **Selected publications.**

- **Joudah, I.**, Müller, S., & Zhu, H. (2025). Air-HOLP: adaptive regularized feature screening for high-dimensional correlated data. *Statistics and Computing*, 35(3), 63.
- Farhood, H., **Joudah, I.**, Beheshti, A., & Müller, S. (2024). Advancing student outcome predictions through generative adversarial networks. *Computers and Education: Artificial Intelligence*, 7, 100293.
- Farhood, H., **Joudah, I.**, Beheshti, A., & Müller, S. (2026). Optimizing cancer classification in high-dimensional data by integrating generative AI with feature selection. *2026 IEEE International Conference on Digital Health (ICDH)*, 359–367.

Full publication list on Google Scholar.

Awards & Recognition

2025 ● **Excellence in Postgraduate Research Award**, Statistical Society of Australia, NSW Branch.

2022 – 2025 ● **Australian Government International Research Training Program (iRTP) Scholarship**, Macquarie University: full tuition (A$96,054) and a living stipend of A$28,854 a year, at 2022 rates.

2018 ● **Second Prize, 9th KFUPM Student Forum**; research project supported by an Undergraduate Student Research Grant.